

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)**B.E. & B.Tech I Sem (Suppl) Examination August 2023****Problem Solving and Programming****(Common to all branches)****Time: 3 Hours****Max Marks: 60****Note: Answer ALL questions from Part-A at one place in the same order and Part-B (Internal Choice)****Part - A****(5Q X 2M = 10 Marks)**

		M	CO	BT
1	List out of various symbols used in flowchart and explain their purpose.	(2)	4	2
2	What are immutable data-types? Give any two examples.	(2)	2	2
3	List various Python selective statements.	(2)	2	4
4	Distinguish between function definition and function invocation.	(2)	3	4
5	What is a file? List the basic file operations.	(2)	6	2

Part - B (50 Marks)**(5Q X 10M = 50 Marks)**

		M	CO	BT
6	(a) Explain the need of OS, compiler, linker and loader.	(5)	1	2
	(b) Design an algorithm for doing arithmetic operations on two numbers and print their results.	(5)	4	3
(OR)				
7	(a) Design a flowchart to find whether a given number is positive or negative.	(5)	4	3
	(b) Explain the evolution of programming languages with appropriate examples.	(5)	1	2
8	(a) How do you compare two strings in python? Explain.	(5)	3	2
	(b) Develop a Python program to implement Binary Search technique.	(5)	3	3
(OR)				
9	(a) Explain Precedence rules. Evaluate the expression $2*3-4/(4+5)*9$.	(5)	2	4
	(b) Develop a python program to find the roots of a quadratic equation (consider all cases).	(5)	3	4
10	(a) What is an Array? How to declare and initialize one-dimensional and two-dimensional arrays? Develop a python program to sort the given set of 'n' integers.	(6)	3	3
	(b) Develop a python program to do matrix addition.	(4)	3	3
(OR)				
11	(a) Develop a python program to store and print elements into an array of size 5 using for loop?	(5)	3	3
	(b) Develop a python program to generate 1 st 10 Fibonacci series.	(5)	3	4

- 12 (a) What is a dictionary? Develop a python program to extract the values associated with a given key from a list of dictionary items. (5) 5 3
- (b) Discuss about various types of function arguments in Python. (5) 3 2
- (OR)
- 13 (a) How do you create a user defined function in Python? Explain with a suitable example. (5) 3 2
- (b) Develop a Python function program to find factorial of a given positive number. (5) 3 4
- 14 (a) Explain various file opening modes in python. (5) 6 2
- (b) Develop a python program to copy contents of one file into another file. (5) 6 3
- (OR)
- 15 (a) Develop a python program to count the number of words and characters of a given file. (5) 6 3
- (b) What is a .csv file? Write a Python script to create *sample.csv* file and insert three records. (5) 6 2
